

Goal

$$Q_{44} := \text{Var}(\Delta a_{ram}[k]) = \frac{2}{1 + \lambda_c} a_x'^2[k] \sigma_x^2[k] \quad (\sigma_x^2 := 4k_B T a_x, \quad a_x' := -a_x K_x / R)$$

Step 1 — closed-loop residual $\varepsilon[k]$ (MA(2))

$$\delta x[k+1] = \lambda_c \delta x[k] - \varepsilon[k]$$

$$\varepsilon[k] = a_x f_T[k] + (1 - \lambda_c) a_x f_T[k-1] + (1 - \lambda_c) a_x f_T[k-2]$$

$\varepsilon[k]$ mixes the current thermal kick with the two most recent ones (controller's $d=2$ -step delay compensation), so consecutive ε 's share raw noise samples — it is *not* white.

Step 2 — autocovariance $\gamma_\varepsilon(\tau)$ of ε

$$\gamma_\varepsilon(0) = \sigma_x^2 \{1 + 2(1 - \lambda_c)^2\}, \quad \gamma_\varepsilon(1) = (1 - \lambda_c)(2 - \lambda_c) \sigma_x^2,$$

$$\gamma_\varepsilon(2) = (1 - \lambda_c) \sigma_x^2, \quad \gamma_\varepsilon(\tau \geq 3) = 0$$

Step 3 — steady-state $\text{Var}(\delta x)$

$$\text{Var}(\delta x) = C_{\delta x} \sigma_x^2, \quad C_{\delta x} = 2 + \frac{1}{1 - \lambda_c^2}$$

Step 4 — lag-1 covariance $\gamma_{\delta x}(1)$

$$\gamma_{\delta x}(1) = E[\delta x[k] \delta x[k+1]] = \lambda_c \text{Var}(\delta x) - E[\delta x[k] \varepsilon[k]]$$

Expand $\delta x[k] = -\sum_{i=0}^{\infty} \lambda_c^i \varepsilon[k-1-i]$ and use $\gamma_\varepsilon(\tau \geq 3) = 0$:

$$E[\delta x[k] \varepsilon[k]] = -\gamma_\varepsilon(1) - \lambda_c \gamma_\varepsilon(2) = -(1 - \lambda_c) \sigma_x^2 \{(2 - \lambda_c) + \lambda_c\} = -2(1 - \lambda_c) \sigma_x^2$$

$$\gamma_{\delta x}(1) = \lambda_c C_{\delta x} \sigma_x^2 + 2(1 - \lambda_c) \sigma_x^2$$

Step 5 — lag-1 autocorrelation ρ_1

$$\rho_1 := \frac{\gamma_{\delta x}(1)}{\text{Var}(\delta x)} = \lambda_c + \frac{2(1 - \lambda_c)}{C_{\delta x}}$$

For $\lambda_c = 0.7$: $C_{\delta x} = 3.961$, $\rho_1 = 0.7 + 0.6/3.961 \approx 0.8515$.

Step 6 — increment-variance identity (general, any stationary signal)

$$\text{Var}(\Delta x_{ram}) = 2 \text{Var}(x_{ram}) - 2 \text{Cov}(x_{ram}[k+1], x_{ram}[k]) = 2(1 - \rho_1) \text{Var}(x_{ram})$$

with $x_{ram} := -\delta x$, so $\text{Var}(x_{ram}) = \text{Var}(\delta x) = C_{\delta x} \sigma_x^2$.

Step 7 — propagate through $a_{ram} = a'_x x_{ram}$

$$\begin{aligned} \text{Var}(\Delta a_{ram}) &= a_x'^2 \cdot \text{Var}(\Delta x_{ram}) = a_x'^2 \cdot \underbrace{2(1 - \rho_1) C_{\delta x}}_{= 2/(1 + \lambda_c)} \sigma_x^2 \\ &= 2/(1 + \lambda_c) a_x'^2 \sigma_x^2 \end{aligned}$$

$C_{\delta x}$ cancels exactly because $(1 - \rho_1) = 1/[(1 + \lambda_c)C_{\delta x}]$ (from Step 5).

Result — Q_{44} (RW mode)

$$Q_{44} = \text{Var}(\Delta a_{ram}) = \frac{2}{1 + \lambda_c} a_x'^2 \sigma_x^2$$

Code: `var_da_inc_factor = 2/(1+lambda_c)` (`motion_control_law_eq17_4state.m:141`), applied as `var_da_inc_factor * (a_hat*K_h/R)^2 * sigma2_dh` (line 444/448).